nano sum.sh

```
#!/bin/bash
sum=0
for ((i=1; i<=10; i++))
do
  sum=$((sum + i))
done
echo "Sum = $sum"
```

chmod +x sum.sh

./sum.sh

🔷 Basic Regex Symbols

Symbol	Meaning
.	Any single character
*	Zero or more occurrences
^	Start of line
$	End of line
[ ]	Range of characters
[^]	NOT in range
\	Escape character

grep "a.c" file.txt. (Matches any one character.

grep "ab*" file.txt (Matches zero or more occurrences.

grep "^MCA" file.txt ((Start of Line)

grep "Lab$" file.txt (END OF LINE)

grep "[0-9]" file.txt

grep "[A-Z]" file.txt

grep "[^0-9]" file.txt (Matches non-digit characters.)

grep -E "a{2}" file.txt (Multiple Occurrences)

grep -E "MCA|BCA" file.txt (Matches either MCA or BCA.)

grep "^[0-9]*$" file.txt (ONLY NUMBER)

grep "^[A-Za-z]*$" file.txt

grep -E "[a-zA-Z0-9]+@[a-zA-Z]+\.[a-z]+" file.txt

DEMO;-

echo "MCA 2026" > test.txt

echo "BCA 2025" >> test.txt

echo "MBA 2024" >> test.txt

grep "^M" test.txt

SED:-

sed 'command' filename

sed 's/old/new/' file.txt (**Replace first occurrence in each line)**

sed 's/old/new/g' file.txt (Replace all occurrences.)

sed -i 's/old/new/g' file.txt (To modify file permanently)

sed '2d' file.txt (Delete specific line)

sed '2,4d' file.txt (RANGE IF LINE)

sed -n '2p' file.txt (PRINT)

sed '2i New Line Added' file.txt (Insert before line 2)

sed '2a Appended Line' file.txt (Append Line)

sed 's/[0-9]/X/g' file.txt (Replace digits with X)

DEMO:-

echo "MCA 2026" > test.txt

echo "BCA 2025" >> test.txt

echo "MBA 2024" >> test.txt

sed 's/2026/2030/' test.txt

🔥 Common sed Commands Summary

Command	Meaning
`s/old/new/`	Replace text
`d`	Delete line
`p`	Print line
`i`	Insert
`a`	Append

AWK:-

Example File (data.txt)

```code
MCA 2026 90
BCA 2025 85
MBA 2024 88
```

Print first column:

```bash
awk '{print $1}' data.txt
```

Output:

```code
MCA
BCA
MBA
```

awk '{print}' file.txt

awk '{print $1}' data.txt

awk '{print $1, $3}' data.txt (Print Multiple Columns)

awk '/MCA/ {print}' data.txt (Print lines containing MCA)

awk '$3 > 85 {print $1}' data.txt (Prints course where marks > 85.)

awk -F ":" '{print $1}' file.txt

```
awk '{sum += $3} END {print sum}' data.txt

awk 'BEGIN {print "Start"} {print $1} END {print "End"}' data.txt
```

DEMO :-

```
echo "MCA 90" > marks.txt

echo "BCA 80" >> marks.txt

echo "MBA 85" >> marks.txt


awk '{sum += $2} END {print "Total =", sum}' marks.txt
```